DIVYANSHU SINGH CHAUHAN

Robotics / Autonomy Software Engineer | Perception & SLAM · Planning & Control · Sensor Fusion · C++/Python/ROS2 on Linux

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Professional Summary

Robotics engineer comfortable across the full autonomy stack, with 4+ years of combined professional and academic experience in perception, localization, planning, and control for real robotic systems. Hands on with particle filter SLAM and EKF/UKF estimation, LiDAR and IMU sensor fusion, and motion planning validated in GPS denied indoor environments on real hardware. Brings 2.5 years of production software engineering (Python microservices, CI/CD, automated testing) to building tooling, observability, and data analysis for robotic systems. Ships and debugs software in C++, Python, and ROS2 on Linux, taken from simulation onto hardware.

Technical Skills

- **Languages:** Python, C++, MATLAB, Bash
- **Robotics & Autonomy:** ROS2, Linux, Motion Planning (A*), Particle Filter SLAM, Occupancy Mapping, Sensor Fusion (LiDAR + IMU), Depth & Point Cloud Perception (RealSense), Trajectory Optimization
- **Estimation & Control:** EKF, UKF, Particle Filter, Kalman Filtering, Sensor Calibration, Monte Carlo Analysis; PID, Cascaded Control, MPC, Optimal & Nonlinear Control, Reinforcement Learning (DDPG)
- **Software, Tooling & Simulation:** Git, CI/CD, Code Review (Bitbucket PRs), Unit Testing (pytest), Docker, FastAPI Microservices, Data Pipelines & Analysis; Gazebo, Unity, Hardware in the Loop, working knowledge of CAN

Experience

Graduate Research Assistant (Trajectory Optimization) — **May 2025 – May 2026**
University of Michigan — *Ann Arbor, MI*

- Formulated and solved a constrained autonomous docking maneuver as a three phase trajectory optimization problem with CasADi/IPOPT, then validated it via Monte Carlo sampling, confirming stable solver convergence across the full tested range of initial conditions.
- Built a modular Python pipeline that decoupled the optimization solver, 6 DOF nonlinear dynamics, and EKF/UKF estimation, so planning, dynamics, and estimation could be developed and tested independently.

Cybersecurity Python Developer — **Aug 2021 – July 2024**
Standard Chartered Bank — *Bangalore, India*

- Built and shipped production Python software (FastAPI microservices and SOAR automation in Splunk Phantom) under CI/CD with code review on Bitbucket and pytest unit tests, cutting incident response time by 40% and analyst workload by 30%.
- Owned a supervised ML pipeline end to end for phishing detection, from feature engineering through evaluation and reporting, reaching 83% detection accuracy on production data.

Deep Learning Intern — **May 2020 – Aug 2020**
HyperVerge Inc. — *Bangalore, India*

- Built deep convolutional and GAN based autoencoders in Python to localize tampered image regions and designed a custom Conv2D layer, raising manipulation detection precision from 75% to 90% at 91% overall accuracy.

Projects

BotLab & ArmLab: Indoor Autonomous Navigation & Manipulation — Aug 2024 – Nov 2024 | C/C++, ROS

- Implemented A* motion planning and particle filter SLAM (LiDAR + IMU fusion) in C/C++ for reliable GPS denied indoor multi room navigation on a real differential drive robot, and added forward/inverse kinematics with Intel RealSense depth perception for autonomous pick and place.

Spacecraft ADCS (SPACE 584) — Aug 2025 – Dec 2025 | MATLAB, Raspberry Pi

- Built a real time EKF in MATLAB for a nonlinear attitude model and confirmed estimator consistency through Monte Carlo analysis and 1σ error bounds.
- Performed magnetometer hard and soft iron calibration and ran Hardware in the Loop testing on a Raspberry Pi 5 with a Helmholtz coil to close the simulation to hardware gap, plus an autonomous detumble controller (1 rev/s to <0.5 deg/s).

Cascaded PID Control for a Flexible Inverted Pendulum — Jan 2026 – Apr 2026 | MATLAB/Simulink

- Designed and tuned a cascaded PID controller to stabilize a 6 state resonant flexible plant within hard $\pm 10^\circ$ limits, separating inner and outer loops by $6\times$ in bandwidth with a custom notch filter to suppress structural resonance ($\omega_n = 20$ rad/s).
- Verified absolute stability via LHP pole placement and the Nyquist criterion, and validated robustness against 100 N impulse disturbances and near zero structural damping ($\zeta = 0.0002$).

Model Predictive Control for Cellular Reprogramming — 2026 – Present | MATLAB, Python

- Casting cellular reprogramming as a constrained receding horizon optimal control problem in a University of Michigan lab, building MATLAB and Python simulations to design the cost and constraint formulation and evaluate closed loop convergence over a nonlinear model.

DDPG Reinforcement Learning for Control (B.Tech Thesis) 2020 – 2021 | Python, ROS, Gazebo

- Implemented a DDPG agent in Python for continuous closed loop grasp control of a robotic hand, trained and evaluated end to end in a ROS/Gazebo simulation.

Education

M.S.E. Aerospace Engineering — Autonomous Systems & Control, GPA 3.77/4.00 **2024 – 2026**

University of Michigan, Ann Arbor *Michigan, USA*

Relevant Coursework: Optimal & Nonlinear Control, Trajectory Optimization, State Estimation, Robotics Systems, Learning Based Control

B.Tech. Mechanical Engineering, GPA 8.27/10.00 **2017 – 2021**

Indian Institute of Technology (IIT) Guwahati *Guwahati, India*

Publications

Deep Learning via LSTM Models for COVID-19 Infection Forecasting in India — *PLOS One*.
Neural Anomaly Search on Lunar Reconnaissance Orbiter Camera Images — *AbSciCon 2022*.